

Hilbert’s tenth problem over $\mathbb{Q}$: a reader’s guide

Companion to OpenAI preprint

OAI:Hilberts-tenth-problem-over-the-rational-numbers-October-6-2026.

Independent exposition; all mathematics is the preprint’s.

Status note. The preprint [1] is unrefereed, and no Lean formalization exists. Its main theorem is *conditional* on another unrefereed OpenAI preprint, the pointwise 2-converse [24, Thm. 1.1], which is used in §3 of the preprint; nothing in the established literature is known to replace it. Our independent audit (five reviews, one per part, plus numerical checks) found no error in the preprint itself. This guide is an exposition, not a referee report: none of the audit’s reviewers is a referee in the field, and the audit is evidence, not proof.

How to use this guide. Read §§1–4 (about 25 minutes), then the part of §5 relevant to your expertise, keeping the preprint open. Source pointers have the form 02:143, meaning file sections/02-reduction.tex, line 143, in the build directory of the October 6, 2026 version. Labels such as red:local-comparison are the preprint’s \label names. We sometimes simplify notation; when we do, we say so. Remarks marked (*ours*) are the guide’s own and are not in the preprint.

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1 The theorem and why it is hard

Statement (thm:main, 01:7). *There is no algorithm which, given $f \in \mathbb{Z}[X_1, \dots, X_n]$, with n part of the input, decides whether f has a zero in $\mathbb{Q}^n$.*

The classical route and its barrier. Over $\mathbb{Z}$ the problem is undecidable by Davis–Putnam–Robinson–Matiyasevich [6, 19]. The obvious transfer to $\mathbb{Q}$ would be an *existential* (Diophantine) definition of $\mathbb{Z}$ in $\mathbb{Q}$, or more generally a Diophantine model of $(\mathbb{Z}, +, \cdot)$ in some $\mathbb{Q}^k$. The known definitions have extra quantifiers. Robinson defined $\mathbb{Z}$ in $\mathbb{Q}$ by a first-order formula [32]. Poonen used a formula of shape $\forall^2 \exists^7$ [26]. Koenigsmann gave a universal definition [15]. None of these is existential.

Mazur’s conjecture on the topology of rational points [20] would rule out a Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$. By Cornelissen–Zahidi [5, Thm. 3.6], it would also rule out any Diophantine model of $\mathbb{Z}$ in a power $\mathbb{Q}^k$. So the direct route is widely expected to be impossible.

How the preprint evades it. The preprint proves a *Turing* reduction $\text{H10}(\mathbb{Z}) \leq_T \text{H10}(\mathbb{Q})$. For each nonconstant f it effectively lists finite “tests”. Each test is a finite system of rational-solvability questions, and

$$f \text{ has a zero in } \mathbb{Z}^n \iff \text{every test succeeds} \quad (\text{intro:finite-test-interface, 01:176}).$$

The right side is an infinite conjunction of existential statements over $\mathbb{Q}$. No single Diophantine set is shown to equal $\mathbb{Z}$, and no model of arithmetic is built. The preprint says so explicitly: the construction “gives no single existential definition of $\mathbb{Z}$ in $\mathbb{Q}$ ” (07:295). Our audit agrees that the result does not conflict with Mazur’s conjecture or with Cornelissen–Zahidi.

Lineage. Rank-one elliptic curves as carriers of integer indices go back to Poonen [27, 28]. The number-field analogue, for rings of integers, was recently settled unconditionally by Koymans–Pagano [16] and by Alpöge–Bhargava–Ho–Shnidman [2]. Those results concern finitely generated $\mathbb{Z}$ -algebras, a class that does not contain $\mathbb{Q}$. The new ingredient here is a *height bound*, which is used to show that a nonstandard “solution” passing every test must in fact be standard.

2 The proof in one page

Fix a nonconstant f . Assume every test succeeds, and suppose for contradiction that f has no integer zero.

Step 0: the decision procedure. Integer zeros are recursively enumerable. Given an oracle for $\text{H10}(\mathbb{Q})$, failed tests are recursively enumerable too. Run both searches in parallel. The arithmetic content is that *some test fails whenever f has no integer zero* (01:181, 02:876–887).

Step 1: compactness. If all tests succeed, the following exist (Lemma `red:compactness`): an elementary extension of the many-sorted structure $(\mathbb{Q}, \mathbb{Z}, \mathbb{R}, F, \text{valuations, heights, } \dots)$; a ring R embedded in its rational sort ${}^*\mathbb{Q}$; and a root $\mathbf{a} \in R^n$ with $f(\mathbf{a}) = 0$. The ring R satisfies a recursive theory $\mathcal{T}$, and every element of R satisfies the pole-parity formula Φ .

Step 2: the index map. $\mathcal{T}$ attaches to each $a \in R$ a point T_a on $E : y^2 = x^3 - d^2x$ over $F = \mathbb{Q}(\sqrt{2})$, compatibly with addition, with $T_1 = P$. Here $E(F)$ has rank one and $mE(F) = \mathbb{Z}P$. Transfer then gives

$$\tilde{T}_a = \eta(a)P, \quad \eta : R \rightarrow {}^*\mathbb{Z} \text{ additive, injective, } \eta(1) = 1.$$

The map η need not be multiplicative, so $f(\eta(\mathbf{a}))$ need not vanish. Put

$$\delta = f(\eta(a_1), \dots, \eta(a_n)) \in {}^*\mathbb{Z}, \quad B = \max(1, |\eta(a_j)|), \quad D = \max(1, \deg f).$$

If $\delta = 0$, elementarity gives an ordinary integer zero. So assume $\delta \neq 0$.

Step 3: slopes and contact primes. At each ordinary precision K , $\mathcal{T}$ writes every $A \in R$ as

$$A = \frac{u_1 + u_2 + u_3}{v}, \quad \text{slopes } s_i = (u_i : v) \in \mathbf{P}^1.$$

Each of the thirteen *contact values* v and $L_b(u_i, v)$ is required to be $\pm k r$, with k in a fixed finite set and r a “tested prime” of R . A prime q of ${}^*\mathbb{Z}$ outside the exceptional set is a *contact prime* of s if some $L_b(u, v)$ has odd normalized valuation at q .

Step 4: local comparison at every contact prime. At such a q , some tested r has odd q -valuation. Pole parity and a coarsening of v_q select one of the two tested branches above r . The paired elliptic tests there give

$$v_q(\tilde{a} - \eta(a)) \geq K \quad \text{for all } a \in R \quad (\text{red:index-congruence}).$$

Step 5: one error term for all A . Apply Step 4 to the root $\mathbf{a}$. Since $f(\tilde{\mathbf{a}}) = 0$, we get $v_q(\delta) \geq K$ at every contact prime of every slope. Then fix $K \geq cD$, where c is the exponent of the height estimate. Chain:

$$\begin{aligned} M_K(\tilde{s}_i)^K &\mid \delta, \quad 0 < |\delta| \leq C_f B^D \quad (02:781, 02:809) \\ \implies h(\tilde{s}_i) &\leq H_K M_K(\tilde{s}_i)^c \leq H_K |\delta|^{c/K} \leq H_K C_f^{c/K} B^{cD/K} \ll_f B \quad (02:816\text{--}826) \\ \implies h(\tilde{A}) &\leq \sum_{i=1}^3 h(\tilde{s}_i) + \log 3 \ll_f B \quad \text{for every } A \in R \quad (\text{red:all-ring-height}). \end{aligned}$$

The implied constant is ordinary and independent of A .

Step 6: canonical height. Each $x(T_{a_j})$ equals $(x_0(a_j) + x_1(a_j)\sqrt{2})/x_2(a_j)$, built from three ring elements. So $h(x(\eta(a_j)P)) \ll_f B$. Transfer of $h(x(nP)) = 2\hat{h}(P)n^2 + O(1)$, with $\hat{h}(P) > 0$ [33], then gives

$$2\hat{h}(P)B^2 \leq CB + C' \implies B \ll_f 1.$$

Hence every $\eta(a_j)$ is an ordinary integer n_j . Additivity and injectivity give $a_j = n_j$. So f has an ordinary integer zero, contradicting $\delta \neq 0$.

In words: The tests cannot force $R = \mathbb{Z}$. They can force every element of R to agree q -adically with its elliptic index, at every prime that the height estimate needs. That one fact turns a

single nonzero integer δ into a uniform height bound on R . Quadratic growth of canonical height then leaves no room for nonstandard indices.

The preprint’s own one-figure summary of Steps 3–5 is in `figures/reduction-overview.tex` (input at 02:790).

3 Dependency diagram

The reduction draws on five inputs. Four are arithmetic: §3, §4, §5 and §6 of the preprint. The fifth is logical: compactness and Skolemization, applied in §2.4.

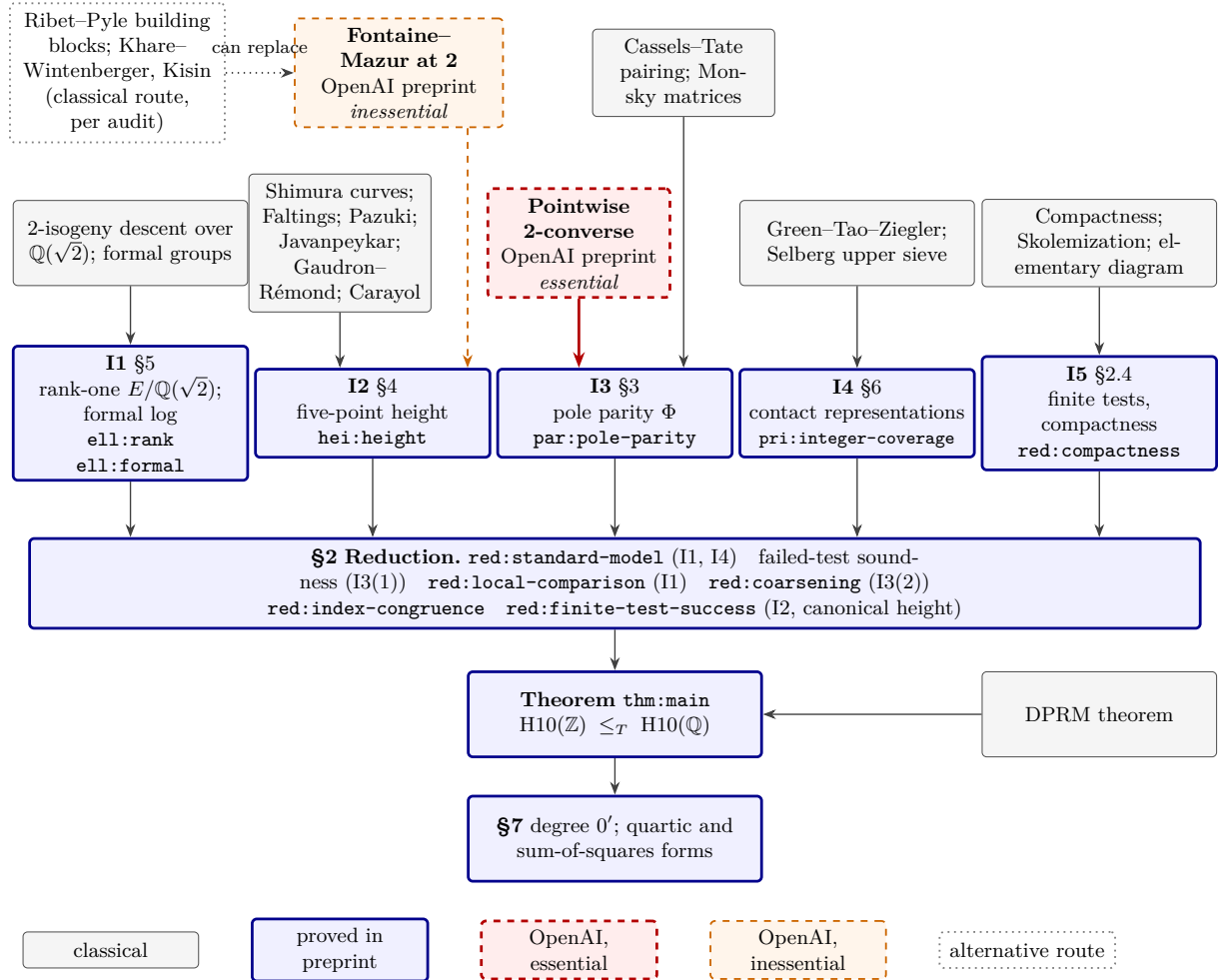


Figure 1: Inputs to the reduction. I3 has two halves. Its completeness half, (1): $\Phi(b)$ holds for every $b \in \mathbb{Z}$; this half needs the pointwise 2-converse. Its soundness half, (2): no odd poles outside S_1 ; this half is elementary (**par:soundness**). Half (1) is used only for the *easy* direction: an integer zero passes every test. Half (2) is used in the hard direction. Not drawn: §6 also uses **ell:formal** from I1, in **pri:integer-tests**.

Reading the diagram.

- **Essential external dependency:** [24, Thm. 1.1] (03:457–462, 03:493). Note where it enters. It proves *completeness* of Φ on $\mathbb{Z}$. Without it, nothing guarantees that an f with an integer zero passes every test, so a failed test would no longer certify “no integer zero”

(02:540–543). The preprint’s Remark (03:528) offers [23] as an alternative. That is also an OpenAI preprint.

- **Inessential external dependency:** [22, Thm. 1.1], used for modularity of r_2 (04:902–954). The audit records a classical replacement: A is a Ribet–Pyle building block, hence a factor of a GL_2 -type abelian variety over $\mathbb{Q}$, hence modular by Khare–Wintenberger and Kisin. Alternatively one can apply Pan’s theorem at some $\ell \geq 5$ and transfer the result via `hei:compatibility`. (These three works are not in the preprint’s bibliography.)
- Everything else is either proved in the preprint or is a published theorem.

4 Notation

Line numbers are approximate pointers to the first definition.

Symbol	Meaning	Where
$F = \mathbb{Q}(\sqrt{2})$, σ	base field; its nontrivial automorphism	<code>ell:data</code> , 02:19
$d = 75 - 53\sqrt{2}$	totally positive, norm 7; (d) is prime with residue field $\mathbb{F}_7$	02:20, 05:11
E , E^σ	$E : y^2 = x^3 - d^2x$ and its conjugate	<code>ell:data</code>
m , P	$mE(F) = \mathbb{Z}P$, with P nontorsion	<code>ell:rank</code> , 02:27
$z = -x/y$, ℓ , ℓ_K	formal parameter ($z(O) = 0$); formal logarithm; its truncation in degree $\leq K$	02:96–102
S_E	primes excluded in the formal estimates	<code>ell:formal</code> , 02:109
R	integral domain of characteristic 0 with $\sqrt{2} \notin \mathrm{Frac} R$; in the proof, the compactness model	02:37, <code>red:compactness</code>
R_F , F_R	$R[\sqrt{2}]$ and $\mathrm{Frac}(R)(\sqrt{2})$	02:40
T_a	point of $E(F_R)$ attached to $a \in R$ (the point axioms)	<code>red:elliptic-axioms</code> , 02:43
$x_0, \dots, y_2$	six ring-valued functions giving the coordinates of T_a	<code>red:coordinate-functions</code> , 02:54
ordinary, $*$, $\sim$	element of the original structure; sort of the elementary extension; image of an element of R	02:63–73
$*\mathbb{Z}$, $*\mathbb{Q}$, $*F$	ambient integer, rational and F sorts	02:63
η	$R \rightarrow *\mathbb{Z}$ with $\tilde{T}_a = \eta(a)P$; additive and injective	<code>red:eta</code> , 02:78
$\mathcal{C}$	five points of $\mathbf{P}^1(\mathbb{Q})$, including ∞ : the branch values of $X \rightarrow X^*$	<code>hei:height</code> ; 04:165–172
$L_b(U, V)$	primitive integral linear form vanishing at $b \in \mathcal{C}$, with $L_\infty = V$ (the “contact forms”)	02:236
contact order	$v_q(L_b(u, v)) - \min\{v_q(u), v_q(v)\}$	<code>hei:radical</code>
$M_S(s)$	product, without repetition, of $q \notin S$ at which some contact order of s is odd (the <i>contact primes</i>)	02:238–243
S , H , c	exceptional set and constants in $h(s) \leq HM_S(s)^c$	<code>hei:height</code>
K	ordinary precision; the final proof uses $K \geq cD$	02:349, 02:783
S_K , M_K , H_K	exceptional set enlarged at precision K ; the corresponding radical and constant	02:597–614
$\mathcal{K}$	finite multiplier set, fixed before everything else	<code>pri:pattern</code> , 06:21
u_i , v , s_i	$u_3 = Av - u_1 - u_2$; slopes $s_i = (u_i : v)$	02:349–362
r , B_r , $\mathfrak{p}_\pm$	tested element; $B_r^2 \equiv 2 \pmod{r}$; the two branch ideals of R_F above r	<code>red:branch-ideals</code> , 02:365–379
h , j_0	auxiliary nonzero ring elements at a branch, <i>independent of a</i>	02:381–399

Symbol	Meaning	Where
$D' \in r^i(R_F)_{\mathfrak{p}}$	ring-language localization membership	red:localization-definition, 02:425
$\mathcal{T}$	the recursive theory	02:335–433
Φ, S_1	pole-parity formula and its exceptional set	par:pole-parity, 02:477; 03:50–61
E_l	$y^2 = x(x-l)(x+3l)$	par:curve, 03:47
$\mathcal{M}, \mathcal{D}_m,$ b', U, H', t, e β, θ, B, J (§3)	multiplier lists and auxiliary quantities in Φ	par:auxiliary-functions
	squarefree representatives of the classes of b', H' ; their supports	03:163–168
K, N, s, λ, D_w (§3)	Monsky-type matrices over $\mathbb{F}_2$	par:matrices, 03:274
δ, B, D, C_f	$f(\eta(\mathbf{a}))$; $\max(1, \eta(a_j))$; $\max(1, \deg f)$; the constant in $ \delta \leq C_f B^D$	02:766–781
J (§2.5), v_*, w_*	convex subgroup generated by negative ring valuations; place above q ; its coarsening	02:657–680
$X, \mathcal{W}, X^*$	Shimura curve of discriminant 210 (genus 5); Atkin–Lehner group $(\mathbb{Z}/2)^4$; the quotient $X^* \simeq \mathbf{P}^1$	04:65–111
$Y, \pi, \mathcal{A}$	fine-level cover, $\pi : Y \rightarrow \mathbf{P}^1$, principally polarized QM abelian surface family	04:175–247
$B, \mathcal{O}$ (§4)	quaternion algebra of discriminant 210; a maximal order	04:44–47
K_0, L, L', T (§4)	fiber field; field of homomorphisms; kernel field of $\beta _{G_L}$; $T = S \cup \{q \mid M\}$	hei:fiber-field, hei:hom-field, 04:888
$\mu_g, \epsilon, \chi, \beta$ (§4)	conjugation isogenies; sign cocycle; Dirichlet character; splitting cochain	04:710–779
R_ℓ, r_ℓ, f, N	4- and 2-dimensional Galois representations; newform; level	04:865–979
J	modular Jacobian; complement; Zarhin’s	04:1162–1246
$J_1(N), D_0, D_Z$	principally polarized complement	
$h, \widehat{h}, h_F$	absolute log Weil height; canonical height; stable Faltings height	02:228, 02:854, 04:13
Ψ_ν, W_0, W, L, X	divided forms, moduli, cutoff and scale in the prime-pattern count	06:289–353

Overloaded symbols (a refereeing hazard).

- B : the index bound (§2); the quaternion algebra (§4); $\text{Supp } \beta$ (§3); the size threshold in `pri:pattern`.
- β : a square class (§3); a cochain (§4).
- K : the precision; a matrix (§3); a compact open subgroup (§4.1). Also K_0 (a field) and $\mathcal{K}$ (the multiplier set).
- L : a field (§4); the sieve cutoff (§6); the unit parameter in `par:soundness`.
- J : a convex subgroup (§2.5); $\text{Supp } \theta$ (§3); $J_1(N)$ (§4).
- m : the index in $mE(F)$; a multiplier in Φ ; the order of χ .
- d : the curve parameter; $d = m_d b$ in Φ ; d_1, d_2 group orders (§6).
- h : a height; an auxiliary ring element.
- c : the height exponent; the scalars $c(g, h)$; c in $H' = \theta c^2$.
- T : the theory $\mathcal{T}$; a prime set (§4); a vector (§6).

- r : the tested prime; a rank; the number of branch values; the representation r_ℓ .

5 Section-by-section guide

Each subsection gives (a) the main statements, (b) the idea, (c) the key computation or trick, and (d) what a referee should check.

5.1 Preprint §2: finite rational tests (02-reduction.tex, 888 lines)

This is the logical core of the preprint. Read it in full.

(a) **Main statements.** *Inputs stated here and proved later.*

- `ell:rank` (02:27): $\text{rank } E(F) = 1$ (proved in §5).
- `ell:formal` (02:109): the formal estimates (§5).
- `hei:height` (02:232): the five-point height estimate (§4).
- `pri:integer-coverage` (02:294): ordinary contact representations (§6).
- `par:pole-parity` (02:477): pole parity (§3).

Lemma red:local-comparison (02:143). Fix ordinary $K \geq 1$, a prime q of ${}^*\mathbb{Z}$ with $q \notin S_E$ and $q > K + 1$, and a place v_* of *F above q . Suppose there are nonzero $h, j_0 \in R$, independent of a , such that for all $a \in R$ the following hold.

- (i) $\tilde{T}_h, \tilde{T}_{ha}$ reduce to O at v_* , $\ell_K(z(\tilde{T}_h)) \neq 0$, and

$$v_*\left(\frac{\ell_K(z(\tilde{T}_{ha}))}{\ell_K(z(\tilde{T}_h))} - \tilde{a}\right) \geq K.$$

- (ii) $\tilde{T}_{j_0}^\sigma$ and $\tilde{T}_{(h-4)a}^\sigma$ reduce to O on E^σ at the same v_* , and

$$v_*(z(\tilde{T}_{(h-4)a}^\sigma)/z(\tilde{T}_{j_0}^\sigma)) \geq K.$$

Then every $\tilde{a}$ is q -integral and $v_q(\tilde{a} - \eta(a)) \geq K$.

Lemma red:standard-model (02:435): $\mathcal{T}$ is recursive, and $\mathbb{Z}$ with $T_a = aP$ is a model.

Lemma red:compactness (02:545): if every finite test succeeds, then there are an elementary extension, a model $R \models \mathcal{T}$, a root $\mathbf{a} \in R^n$, and an embedding $R \hookrightarrow {}^*\mathbb{Q}$ whose image satisfies Φ .

Lemma red:coarsening (02:643): let $q \notin S_K$ be an ambient prime counted by some $M_K(\tilde{s}_i)$, and r a tested element with $v_q(\tilde{r})$ odd, as in `red:odd-r`. Then some branch $\mathfrak{p} \in \{\mathfrak{p}_\pm\}$ and some place v_* above q satisfy

$$D' \in r^l(R_F)_{\mathfrak{p}} \implies v_*(\tilde{D}') \geq l$$

for all ordinary $l \geq 1$. Moreover, tested-formal points reduce to O at v_* .

Proposition red:index-congruence (02:722): for every q counted by $M_K(\tilde{s}_i)$ and every $a \in R$, the element $\tilde{a}$ is q -integral and $v_q(\tilde{a} - \eta(a)) \geq K$.

Proposition red:finite-test-success (02:755): if all tests for f are solvable, then f has a zero in $\mathbb{Z}^n$.

(b) Idea. The theory $\mathcal{T}$ says, in ring language, everything the integers satisfy that the proof needs:

- the point axioms;
- for each K , a representation axiom: three slopes whose thirteen contact values are multiplier times tested prime;
- at each tested prime, two residue branches carrying the paired elliptic tests.

The tests are the finite fragments of the Skolemized $\mathcal{T} + f(\mathbf{a}) = 0$ evaluated on ground terms. They are answered by rational-solvability queries, with Φ imposed on every ground term (constraints (a)–(d), 02:518–524). Compactness [18] turns “all tests succeed” into a model R inside ${}^*\mathbb{Q}$. Two connections then do the work. An ambient contact prime q is linked to a tested branch by an odd valuation and pole parity. That branch is linked to the index map by the local comparison.

(c) Key tricks.

1. *The “ $h - 4$ ” trick* (proof of `red:local-comparison`, 02:178–218). Test (ii) uses only additivity of η . It gives $v_q(\eta((h - 4)a)) \geq K$, hence

$$V_a := \eta(ha) \equiv 4\eta(a), \quad V_1 \equiv 4 \pmod{q^K}.$$

So V_a/V_1 is q -integral and congruent to $\eta(a)$, because $q \neq 2$. Integrality is exactly the hypothesis that the truncated-log estimate (`ell:truncation`) needs. Test (i) then identifies $\tilde{a}$ with V_a/V_1 to precision K . The same h must serve both a and 1; this is why h, j_0 are chosen independently of a (02:220).

2. *Odd contact forces an odd tested value* (02:623–632). Fix $q \notin S_K$, and let t be the smaller of $v_q(\tilde{u}_i)$ and $v_q(\tilde{v})$. The minimum of the five contact valuations equals t , because some pair of forms has unit determinant. If some contact depth is odd, the five valuations do not all have the same parity, so one is odd. Its factor k is a q -unit, so $v_q(\tilde{r})$ is odd.
3. *Coarsening* (proof of `red:coarsening`, 02:656–717). R may have poles at q . Let $J \subset {}^*\mathbb{Z}$ be the convex subgroup generated by the negative values of v_q on $\tilde{R} \setminus \{0\}$; this is an external construction. Pole parity, applied to b and ab , shows that every ring valuation lying in J is even. So the odd $v_q(\tilde{r})$ lies outside J , and is positive. The coarsened valuation $w_* : {}^*F \rightarrow {}^*\mathbb{Z}/J$ is ≥ 0 on R_F . Its center is a prime of R_F containing r . Since $R_F/rR_F \simeq (R/rR)^2$, the center is exactly $\mathfrak{p}_+$ or $\mathfrak{p}_-$.
4. *One δ for all A* (02:792–840). The internal universal statement “every prime counted by $M_K(\tilde{s}_i)$ divides δ to order $\geq K$ ” is established, and then the transferred finite-product fact gives $M_K(\tilde{s}_i)^K \mid \delta$. The exponent c must be fixed, because $K \geq cD$ is chosen once (02:264–266).

In words: The ring R may be wildly nonstandard. Yet at every prime that matters for heights, it is locally indistinguishable from its own elliptic indices.

(d) What to check.

- *Expressibility* (02:407–433). Every axiom of $\mathcal{T}$ must be first-order in rings: no valuation function, and fixed exponents only. The branch localizations must be defined via R/rR rather than via an ambient valuation.
- *The tests* (02:503–543). Skolemization is effective. The witnesses for Φ are auxiliary and do *not* generate new ground terms. The DNF and sum-of-squares conversion is correct.

- *Soundness of a failed test* (02:540). This uses `red:standard-model` and Proposition `par:pole-parity(1)`; the latter is the conditional input.
- *Many-sorted ambient structure* (02:63–73). Every transferred fact must be a first-order truth of the standard structure with ordinary parameters. In particular: the formal estimates only for integer multiples of P, P^σ , and no infinite series (02:135–140); the canonical-height comparison (02:851–857); and the finite-product divisibility fact (02:802–814).
- *Coarsening*. Check the claims “ $J \neq 0$ implies $J \supseteq \mathbb{Z}$ ” and “a positive class in ${}^*\mathbb{Z}/J$ exceeds every ordinary l ” (02:705–711). Check that reduction to O at the good model is $v_*(x) < 0$ (02:713–716).
- S_K (02:597–604). It must contain the Φ exceptions, the supports of $\mathcal{K}$, the primes needed for unit determinants, the bad primes, and the primes $\leq K + 1$ together with S_E .
- *Constants*. C_f, H_K and $K \geq cD$ are ordinary, and the final constant is independent of A and $\mathbf{a}$ (02:838–840). Check the step $2\hat{h}(P)B^2 \leq CB + C'$ (02:862).
- Our audit’s verdict on this section: valid given its inputs.

5.2 Preprint §3: pole parity (03-parity.tex, 718 lines)

(a) **Main statements.** *Proposition `par:pole-parity`* (02:477). There are a positive existential ring formula $\Phi(b)$ with rational coefficients and a finite set S_1 such that:

- (1) $\mathbb{Q} \models \Phi(b)$ for every $b \in \mathbb{Z}$;
- (2) if $b \in \mathbb{Q}^\times$ satisfies Φ and $q \notin S_1$, then $v_q(b) < 0$ implies $v_q(b) \in 2\mathbb{Z}$.

The formula (03:50–61). $\Phi(b)$ holds if $b = 0$, or if for some multipliers $m \in \mathcal{M}$, $m_d \in \mathcal{D}_m$ and some $t, e \in \mathbb{Q}$ we have

$$d = m_d b, \quad b' = mb, \quad U = b \frac{1 - dt^2}{1 + dt^2}, \quad H' = b'(b' - U^2), \quad 1 + dt^2 \neq 0, \quad eH' \neq 0,$$

and both E_e and $E_{eH'}$ have a rational point with $x \in b'\mathbb{Q}^{\times 2}$. Here $E_l : y^2 = x(x - l)(x + 3l)$.

Supporting lemmas.

- `par:soundness` (03:71): property (2), for any lists.
- `par:cross-squares` (03:170): mutual local squares of β, θ .
- `par:descent` (03:300): under matrix conditions, $\dim_{\mathbb{F}_2} \text{Sel}_2(E_l/\mathbb{Q}) = 3$, and $(\beta, 1)$ is a Selmer class outside the two-torsion classes.
- `par:global-point` (03:464): that class comes from a rational point.
- `par:simultaneous-matrices` (03:543): one n works for both $l = n$ and $l = n\theta$.

(b) **Idea.** *Soundness.* If $v_q(b) = k < 0$ is odd, then $v_q(U) = k$ and $v_q(H') = 3k$. So e and eH' have opposite valuation parities. On the twist whose parameter has even valuation, rescale to a unit parameter. There every nonzero x has even valuation, which contradicts $x \in b'\mathbb{Q}_q^{\times 2}$.

Completeness. For an integer b , choose the multipliers and t by weak approximation, so that b' and H' are local squares at each other’s odd-support primes. Then build a squarefree n for which both E_n and $E_{n\theta}$ have 2-Selmer rank exactly one beyond torsion, with $(\beta, 1)$ as the extra class. The 2-converse and Cassels–Tate then promote the Selmer class to a point.

(c) **Key computations.**

- *Selmer equations.* In normalized coordinates $(3^\delta \prod q_i^{u_i}, \prod q_i^{v_i})$ (03:384), the local conditions read

$$Ku + D_\lambda v = \delta \lambda, \quad Nv = 0, \quad (\lambda + s)^t v = 0,$$

with $K + K^t = ss^t + D_s$ from quadratic reciprocity (03:282). The matrices follow Monsky’s viewpoint [13, Appendix]. The target $\ker K = \langle \mathbf{1}, \mathbf{1}_B \rangle$ and $\lambda \notin \text{im } K$ force $v = 0$, $\delta = 0$, $u \in \ker K$. Hence there are at most 8 classes, and $\tau_1, \tau_2, (\beta, 1)$ attain the bound.

- *The converse step* (03:470–514). By $r + \dim \text{Sha}[2] = 1$, the 2^∞ -Selmer corank is ≤ 1 . The pointwise 2-converse (hypotheses $C(\mathbb{Q})[2] \neq 0$ and $\text{corank} \in \{0, 1\}$) gives Sha finite. Cassels–Tate on the finite 2-primary part then makes $\dim \text{Sha}[2]$ even, so $\text{Sha}[2] = 0$ and $r = 1$.
- *Simultaneous matrices* (03:585–704). The two kernel targets reduce to $K\mathbf{1}_B = 0$ plus nonsingularity of the deleted matrix of $K^\# = K + \lambda a^t$. The B – I block is built from a basis of the even-sum hyperplane. Adding J toggles diagonal entries independently, and a nonzero multilinear polynomial over $\mathbb{F}_2$ has a non-root.

In words: Odd poles are detected by a sign-type invariant of two quadratic twists. Integers pass because one can always choose a twist pair whose Mordell–Weil groups contain the needed square class. “Always” relies on a converse theorem that rules out a Selmer class coming from an infinite Sha.

(d) **What to check.**

- Positivity of Φ : the disjunction over multipliers is finite, and $\neq$ is encoded as $az = 1$ (03:63–69).
- The valuation case analysis in `par:soundness`, including $t = 0$ and $q \neq 3$ (03:76–103). The audit tested soundness on 18k random q -adic cases.
- The local Kummer images at $q \nmid 6l$, at $q \mid l$, at 3, at 2 (necessity only), and at ∞ (03:328–390).
- Exact hypotheses of [24, Thm. 1.1] as quoted (03:457–462): full 2^∞ -Selmer corank, not just Sel_2 . The family E_l is non-CM, with additive reduction at every $q \mid l$.
- Cassels–Tate alternation for the principal polarization (03:496–514). The locator [29, Cor. 9] is unverified; it is not load-bearing, since Cassels [4] suffices.
- The Dirichlet realization of prescribed Legendre symbols with reciprocity-consistent reverses (03:692–703).

5.3 Preprint §4: the five-point height estimate (04-height.tex, 1326 lines)

(a) **Main statement.** *Theorem hei:height* (02:232). There are five distinct points $\mathcal{C} \subset \mathbf{P}^1(\mathbb{Q})$, including ∞ , a finite set S , and constants $H, c > 0$ with the following property. For each $s = (u : v) \in \mathbf{P}^1(\mathbb{Q}) \setminus \mathcal{C}$,

$$h(s) \leq H M_S(s)^c,$$

where $M_S(s)$ is the product of the primes $q \notin S$ at which some $v_q(L_b(u, v)) - \min\{v_q(u), v_q(v)\}$ is odd. *Corollary hei:enlarge-exceptions* (02:270): enlarging S changes H but not c .

In words: If a rational point of $\mathbf{P}^1$ meets the five points to odd order only at a few primes, its height is polynomially bounded in the product of those primes. In terms of the naive height

$e^{h(s)}$, this is an exponential-in-a-power bound; the audit calls it “exponential-abc strength, not abc strength”. The proof uses a finiteness statement that need not be effective (04:680).

Intermediate results.

- **hei:fiber-field** (04:303): the fiber field K_0 has bounded degree, is unramified outside $S \cup \{q \mid M\}$, and fibers have good reduction outside S .
- **hei:base-height** (04:364): $h(\pi(y)) \leq C(1 + \max\{0, h_F(\mathcal{A}_y)\})$.
- **hei:hom-field** (04:584): a Galois field L of bounded degree, unramified outside T , containing all homomorphisms between conjugates.
- **hei:sign-splitting** (04:766): a character χ of order $\leq CM$ and a cochain β with $\beta^2 = \chi$ and $\partial\beta = \epsilon$.
- **hei:compatibility** (04:986).
- **hei:level-bound** (04:1043): $N \leq CM^a$, with $a = 2$ allowed.
- **hei:modular-factor** (04:1170): $J_1(N) \sim_{L'} A \times D_0$.
- Final bound (**hei:faltings-final**, 04:1293): $h_F(A) \leq CM^C$.

(b) Idea.

1. Take the Shimura curve X of the indefinite quaternion algebra of discriminant 210. It has genus 5. Its full Atkin–Lehner quotient X^* is $\mathbf{P}^1$ over $\mathbb{Q}$, with exactly five branch values, all rational; these form $\mathcal{C}$.
2. At fine level $Y \rightarrow X$ there is a principally polarized QM abelian surface family $\mathcal{A} \rightarrow Y$. Fibers over the same point of X^* are isogenous.
3. For rational s , the local equation $t = z^2$ shows that only odd contact ramifies the fiber field (**hei:fiber-field**).
4. A theta-height comparison (**hei:base-height**) shows that $h(s)$ is controlled by the Faltings height of a fiber A .
5. If the centralizer of B in $\text{End}^0(A)$ is an imaginary quadratic field, then A is isogenous to E^2 with E CM, and its height is uniformly bounded. Otherwise $\text{End}^0 = B$, and A is isogenous to all its Galois conjugates. The descent obstruction is a ± 1 cocycle, split by a cochain β with controlled order and inertia. This yields an odd two-dimensional 2-adic representation r_2 , de Rham with Hodge–Tate exponents 0, 1.
6. Modularity gives a weight-2 newform of level $N \ll M^2$, and A is a factor of $J_1(N)$ over a field of degree $\ll M$.
7. Javanpeykar bounds $h_F(J_1(N)) \ll N^{12}$. The Gaudron–Rémond isogeny theorem, with its explicit dependence on the dimension, transfers this to $h_F(A) \ll M^C$.

The final step follows von Känel’s modular height strategy [34]. The main external tools are Pazuki’s theta height [25], Javanpeykar’s bound for Belyi curves [14], the Gaudron–Rémond isogeny theorem [8], Faltings’ theorems [7], and Carayol’s local–global compatibility [3]. The descent step follows the theory of building blocks [31, 30].

(c) Key tricks.

- *Parity from ramification index two* (04:312–346). After an unramified extension the local equation is $t = z^2$. Even contact gives unramified lifts.

- *Normalize degrees, not isogenies* (04:729–756). With $a_g = (\deg \mu_g)^{-1/4}$, the discrepancy is a pure sign, so no bound on $\deg \mu_g$ is needed.
- *Cheap splitting* (04:781–851). Choose an odd character modulo each odd prime of T where the Brauer class is nonzero. The order is then $\leq 2 \prod_{q \in T, q \neq 2} (q-1) \leq CM$. A quadratic twist removes inertia at primes outside T .
- *Level bound* (04:1062–1146). At large $q \in T$ the representation is tame, so the exponent is ≤ 2 . At the finitely many small primes, a uniform upper-ramification cut (from bounded-degree local fields and the semistable reduction criterion) gives a uniform Swan bound. At $q = 2$ the conductor is read from r_3 , via `hei:compatibility`.

(d) What to check.

- *Geometry* (04:85–172). No elliptic points (because $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$ do not embed in B). Area 8 gives $g = 5$. Then Riemann–Hurwitz, and five distinct labels 30, 42, 70, 105, 210 (see §6.4). Cross-check against Long–Maclachlan–Reid (signature $(0; 2^5; 0)$) and Nualart Riera [17, 21].
- The symplectic embedding and the choice of level (04:199–247).
- The theta-height argument and its uniformity in the field of definition (04:372–549). This is the most technical analytic step.
- The CM case: non-effective but uniform (04:658–680).
- The hypotheses of the modularity input (04:902–954): odd, irreducible, de Rham with distinct weights, finitely ramified. If you prefer, check the classical replacement instead (see §3).
- The conductor exponent at small primes. The citation of the semistability criterion (04:1115) is better given as SGA7 IX 4.7, per the audit.
- The arithmetic N^{12} (from $\deg_B \leq CN^2$ and $g \leq CN^2$), and the bound $\log \deg \phi \leq CM^C$ from `hei:gr` (04:1207–1295).
- The audit found no error here, and no cheap counterexample family.

5.4 Preprint §5: the rank-one curve (05-elliptic.tex, 189 lines)

(a) Statements. `ell:rank`: $\text{rank } E(F) = 1$; fix m, P with $mE(F) = \mathbb{Z}P$.

`ell:formal`: there is a finite set S_E with the following property. Let $p \notin S_E$, let $v \mid p$ with $v(p) = 1$, and let $nP, n'P$ reduce to O with $n' \neq 0$. Then

$$v\left(\frac{z(nP)}{z(n'P)}\right) = v_p(n/n').$$

If moreover $p > K + 1$ and $v_p(n/n') \geq 0$, then $\ell_K(z(n'P)) \neq 0$ and

$$v\left(\frac{\ell_K(z(nP))}{\ell_K(z(n'P))} - \frac{n}{n'}\right) \geq K.$$

The same holds for E^σ, P^σ .

(b) Idea. The rank comes from a 2-isogeny descent over $\mathbb{Q}(\sqrt{2})$, which is possible because $\mathbb{Z}[\sqrt{2}]$ is a PID and totally positive units are squares. One finds $|\text{im } \alpha| = 8$ and $|\text{im } \alpha'| = 1$, hence $4 \cdot 2^{\text{rank}} = 8$. The formal statements come from the logarithm being valuation-preserving on the formal kernel when $p > 2$.

(c) **Key computation.** For $t > K$ and $p > K + 1$ one has $t - 1 - v_p(t) \geq K$ (`ell:tail-exponent`). Hence $v(\ell(Z) - \ell_K(Z)) \geq v(Z) + K$ whenever $v(Z) \geq 1$ (`ell:tail`), and the truncation error is negligible.

(d) **What to check.** The explicit point Q and the sign argument (05:67–75). The computation at $\mathfrak{d}$ using $-4 = 3$, a nonsquare in $\mathbb{F}_7$ (05:84–91). The tail inequality, including the edge case $p = K + 2$. The audit verified §5 by exact computation, including a 400-digit r -adic test at $r = 17, 23$.

5.5 Preprint §6: prime patterns (06-primes.tex, 705 lines)

(a) **Statements.** *Lemma pri:pattern* (06:21). There is a finite $\mathcal{K} \subset \mathbb{Z}_{>0}$ with the following property. For every $A \in \mathbb{Z}$ and every real $B > 0$ there are integers u_1, u_2, v , with $v \neq 0$ and $u_3 = Av - u_1 - u_2$, such that each of the thirteen numbers v and $L_b(u_j, v)$ ($b \neq \infty$) equals $\pm kr$ with $k \in \mathcal{K}$ and $r > B$ prime. Each such r splits in F , is good for E and E^σ , and at either place above r

$$\gcd(d_1, d_2) \mid 4, \quad r \nmid d_1 d_2,$$

where $d_1, d_2 = \#\tilde{E}(\mathbb{F}_r), \#\tilde{E}^\sigma(\mathbb{F}_r)$.

Corollary pri:integer-tests (06:53). At each place above such an r (large enough for precision K), take $j_0 = d_2$ and

$$h \equiv 0 \pmod{d_1}, \quad h \equiv 4 \pmod{d_2 r^K}.$$

Then conditions (i) and (ii) of `red:local-comparison` hold in $\mathbb{Z}$ for all a . Combining the two results gives *Corollary pri:integer-coverage*.

(b) **Idea.** Take $r \equiv 1 \pmod{8}$ and $r \equiv 3 \pmod{7}$. Then r splits in F , and $7 = dd^\sigma$ is a nonsquare mod r . So E and E^σ reduce to opposite twists of $y^2 = x^3 - x$, with orders $r + 1 \pm 2a$ where $a^2 + b^2 = r$. The 2-part of the gcd is exactly 4. An odd common factor l needs $l \mid a$ and $b^2 \equiv -1 \pmod{l}$. Small such l are excluded by congruences. Large l are removed by an upper sieve. The lower bound for thirteen simultaneous prime values of linear forms in (u_1, u_2, v) comes from Green–Tao–Ziegler [9, 10, 11].

(c) **Key tricks.**

- *Order of choices* (06:121–125): first $\mathcal{K}$, then A , a box and a congruence class, then the cutoff L , and finally the scale X . $\mathcal{K}$ must be independent of A , because it appears in the single axiom scheme of $\mathcal{T}$.
- *Lower versus upper bounds.* GTZ gives $N_{A,L}(X) \geq c_A (\log L)^{-C_0} X^3 / (\log X)^{13}$ (`pri:lower`). The patterns with a bad $l > L$ number at most $C_A X^3 / (L (\log X)^{13}) + o(\cdot)$ (`pri:total-loss`). Choose L large. There is no parity barrier, because the sieve is used only for upper bounds.
- (*ours*) Why three slopes rather than one? If $A = u/v$, then $L_b(Av, v) = v L_b(A, 1)$ is a multiple of v . Such values could not all be $\pm k \cdot (\text{large prime})$ alongside v itself. Three summands give three free variables, and thirteen forms with pairwise non-proportional linear parts (06:186–205), which is the finite-complexity setting of GTZ.

(d) **What to check.**

- Pairwise independence of the thirteen forms modulo p for p above a threshold independent of A (06:198–205).

- The catalogue construction of $\mathcal{K}$ (06:223–250).
- The local factors β_p (06:392–406) and the negligibility of prime powers.
- Uniformity of the two upper sieves in L , r and l (06:550–668).
- The cited form of GTZ (06:355–378). The audit suggests adding Manners or Leng–Sah–Sawhney as a backstop to the blog-hosted 2024 erratum [12].

5.6 Preprint §7: Turing degree and normal forms (07-consequences.tex, 108 lines)

(a) Statement (thm:degree-normal-form, 07:197). $H10(\mathbb{Q})$ has Turing degree $0'$. So do two restricted problems: total degree ≤ 4 , and $\sum_i q_i^2$ with $\deg q_i \leq 2$ given as a list. No bound is placed on the number of variables or on coefficients.

(b,c) Idea. The parallel search of §2 with an oracle gives $H10(\mathbb{Z}) \leq_T H10(\mathbb{Q})$. The fixed data are independent of f , so there is one oracle machine. Both sets are c.e. An arithmetic circuit with one new variable per gate gives many-one reductions $H10(\mathbb{Q}) \leq_m S_2 \leq_m H_{\leq 4}$.

(d) What to check. That the fixed data are hardcoded, not searched for (02:461–467, 07:226–228). The preprint itself notes that many-one completeness of $H10(\mathbb{Q})$ is *not* claimed (07:289–296).

6 Worked toy illustrations

6.1 Why the point axioms exclude $\mathbb{Z}[1/2]$, but not nonstandard denominators (*ours*)

Take $R = \mathbb{Z}[1/2] \subset \mathbb{Q}$ in the standard structure, embedded by the identity, as in the setting of 02:63–78. Then $F_R = \mathbb{Q}(\sqrt{2}) = F$. The axioms demand $T_{1/2} \in mE(F) = \mathbb{Z}P$, say $T_{1/2} = nP$. Additivity gives $P = T_1 = 2T_{1/2} = 2nP$. Since P has infinite order, $2n = 1$, which is impossible.

The same argument works for any standard rational $1/k$ with $k > 1$, since $k\eta(1/k) = \eta(1) = 1$ has no solution in ${}^*\mathbb{Z}$. So a model of the point axioms contains no standard non-integral rationals.

The argument says nothing about an element such as $1/N$ with $N \in {}^*\mathbb{Z}$ nonstandard. Then N need not lie in R , and additivity only constrains $\eta(k/N) = k\eta(1/N)$ for standard k . This is why the preprint needs the rest of its machinery: local comparisons, height bounds, and canonical height growth. The condition $T_a \in mE(F_R)$, rather than $T_a \in E(F_R)$, is what makes η land in ${}^*\mathbb{Z}$ (02:75–78).

6.2 The rank-one curve over $\mathbb{Q}(\sqrt{2})$

The data are $d = 75 - 53\sqrt{2}$, with $N(d) = 75^2 - 2 \cdot 53^2 = 7$, and $d > 0$ at both real embeddings. The preprint gives (05:67–70)

$$w = -5 + 4\sqrt{2} = d(1 + \sqrt{2})^3, \quad w^3 - w = -8d, \quad Q = (-dw, 2\sqrt{2}d^2) \in E(F).$$

To check this: $(1 + \sqrt{2})^3 = 7 + 5\sqrt{2}$, and $(75 - 53\sqrt{2})(7 + 5\sqrt{2}) = -5 + 4\sqrt{2}$. Also $w^3 = -605 + 428\sqrt{2}$, so $w^3 - w = -600 + 424\sqrt{2} = -8d$. Finally $x^3 - d^2x = -d^3(w^3 - w) = 8d^4 = y^2$.

The class of $-dw$ has signs $(-, +)$ at the two real embeddings. The torsion classes $1, -1, d, -d$ have equal signs at both. So $|\text{im } \alpha| = 8$. Together with $\text{im } \alpha' = 1$ and `ell:isogeny-index`, this gives $8 = 4 \cdot 2^{\text{rank}}$, so the rank is 1.

6.3 Integer witnesses at a small split prime (*ours, computed*)

Take $r = 17$. Then $17 \equiv 1 \pmod{8}$ and $17 \equiv 3 \pmod{7}$, as in `pri:prime-residues`. The square roots of 2 mod 17 are 6 and 11. At the place $\sqrt{2} \mapsto 6$ we have $d \mapsto 12$ and $d^\sigma \mapsto 2$, with $12 \cdot 2 \equiv 7$. A direct point count gives

$$d_1 = \#\tilde{E}(\mathbb{F}_{17}) = 20, \quad d_2 = \#\tilde{E}^\sigma(\mathbb{F}_{17}) = 16.$$

This agrees with $r + 1 \pm 2a$ for $17 = 1^2 + 4^2$, $a = \pm 1$. Here $\gcd(20, 16) = 4$ and $17 \nmid d_1 d_2$.

For $K = 1$, Corollary `pri:integer-tests` asks for $h \equiv 0 \pmod{20}$ and $h \equiv 4 \pmod{16 \cdot 17}$. Take $h = 820$ and $j_0 = 16$. Then hP and haP lie in the formal kernel of E , and $(h - 4)a = 816a = 3 \cdot 16 \cdot 17 \cdot a$. So $v_{17}((h - 4)a/d_2) \geq 1 = K$, as test (ii) requires.

At the other place the roles of the two curves swap: $d_1 = 16$, $d_2 = 20$, $h = 1024$, $j_0 = 20$. The lemma itself uses only primes above a large threshold; $r = 17$ serves only as illustration.

6.4 The genus-5 cover and its five branch values

The Eichler area formula gives $\frac{1}{6} \prod_{p|210} (p - 1) = \frac{1}{6} \cdot 1 \cdot 2 \cdot 4 \cdot 6 = 8 = 2g - 2$, so $g(X) = 5$ (04:93–97).

$\mathcal{W} \simeq (\mathbb{Z}/2)^4$ has order 16, and every nontrivial stabilizer has order 2. Riemann–Hurwitz reads

$$8 = 16(2g(X^*) - 2) + r \cdot 8 \iff 2g(X^*) - 2 + \frac{r}{2} = \frac{1}{2},$$

so $r = 5 - 4g(X^*)$.

Fixed points of w_m for $m = 30, 42, 70, 105, 210$ exist because $\mathbb{Q}(\sqrt{-m})$ embeds in B . For example, for $m = 30$ the only prime of 210 not dividing m is 7, and $-30 \equiv 5$ is a nonsquare mod 7 (04:139–146). So $r \geq 5$, which forces $g(X^*) = 0$ and $r = 5$. Each branch value is rational because its label is Galois-stable.

Note that the five points $\mathcal{C}$ are not written down explicitly in the preprint. Only ∞ is normalized.

6.5 Pole parity at $b = 1/q$

Take $q \notin S_1$ and $b = 1/q$. For any witnesses, $v_q(d) = v_q(b') = -1$ and $v_q(U) = -1$, so $v_q(H') = -3$. One of e and eH' , call it l , has even valuation $2a$. On $Y^2 = X(X - L)(X + 3L)$ with $L = q^{-2a}l$ a unit, every nonzero X has even valuation. But the required x lies in $b'\mathbb{Q}^{\times 2}$, which has odd valuation. So $\Phi(1/q)$ fails.

The argument gives no obstruction for $b = 1/q^2$, and `par:pole-parity(2)` does not exclude even poles (02:488).

6.6 What the height estimate says when $M = 1$ (*ours*)

If every contact order of s outside S is even, then $M_S(s) = 1$ and $h(s) \leq H$. So only finitely many such s exist. In the reduction, $M_K(\tilde{s})^K \mid \delta$ means that the contact radical is at most $|\delta|^{1/K}$, a “ K -th root” of a single fixed error term. This is why the exponent c must be independent of K (02:259–266).

7 Audit status and open points

What was checked (from ../AUDIT.md).

Part	Finding
§2 and §7	Valid given the inputs. The test conditions are ring-existential. The transferred facts are first-order. The choice $K \geq cD$ and the step $B^2 \ll B$ are correct. Turing reduction, not a Diophantine model.
§3	No error. Soundness was tested on 18k random q -adic cases. $\dim \text{Sel}_2 = 3$ was checked on 9 values of l . 480 randomized matrix trials passed. Completeness depends on [24].
§4	No error. Follows von Känel’s method. Level $N \ll M^2$. Exponential-abc strength. No cheap counterexample family was found.
§5	Verified by exact computation; 400-digit r -adic test at $r = 17, 23$, including $p = K + 2$.
§6	Correct. GTZ for 13 fixed forms in 3 variables, so no parity barrier. The sieve bounds are uniform as claimed.

What is conditional. The theorem depends on [24, Thm. 1.1]. Known 2-converse results cover CM families or special Heegner families. The family here is non-CM, with additive reduction at every prime dividing l . The alternative [23] is also an OpenAI preprint. The audit of the pointwise 2-converse found no error, with sketched infrastructure gaps.

The Fontaine–Mazur input [22] is cited but not needed. The audit recommends citing the classical route, noting that $\ell = 2$ is the hardest choice since B is ramified at 2. The Fontaine–Mazur preprint itself needs expert review of three points: characteristic-2 patching, marked-line connectedness, and classicality transport (which relies on another unrefereed preprint).

Minor fixes suggested by the audit.

1. 01:183: “elementary extension of the rational field” should read “of the many-sorted structure”. §2 states this correctly.
2. §6: add Manners or Leng–Sah–Sawhney as a backstop to the blog-hosted GTZ erratum.
3. 04:1115: cite the semistability criterion to SGA7 IX 4.7.
4. §3: the locator “Poonen–Stoll, Corollary 9” is unverified. It is not load-bearing.

Suggested priorities for a referee (*ours*).

- §2, and the logic in particular: an expert in model theory.
- §4: an arithmetic geometer. This is the most technical part, and the theta-height and level-bound steps are where most of the length goes.
- The pointwise 2-converse: separate expert review. In the audit’s words, it is “the theorem’s real foundation”.

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